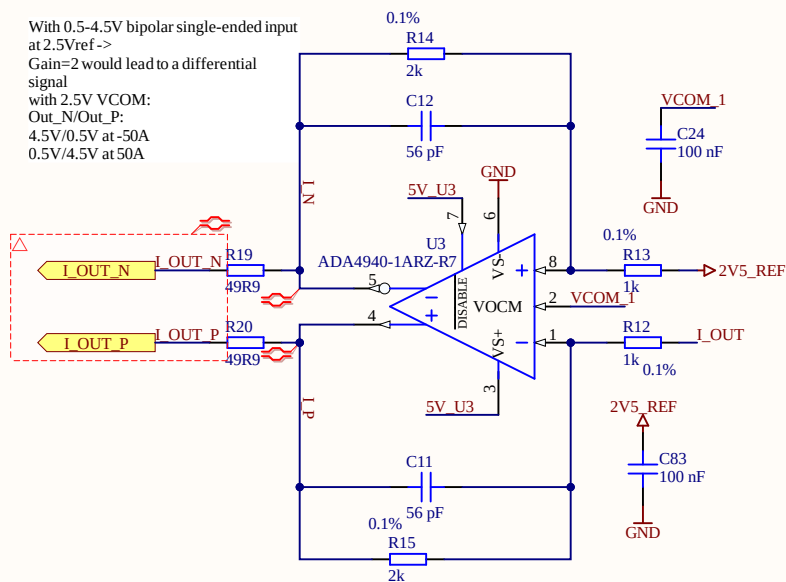
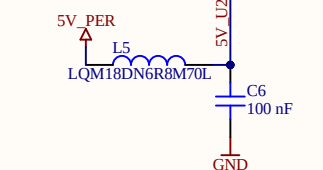


With 0.5-4.5V bipolar single-ended input at 2.5Vref ->
Gain=2 would lead to a differential signal with 2.5V VCOM:
Out_N/Out_P:
4.5V/0.5V at -50A
0.5V/4.5V at 50A

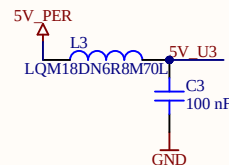


Place close to U2 OPAMP

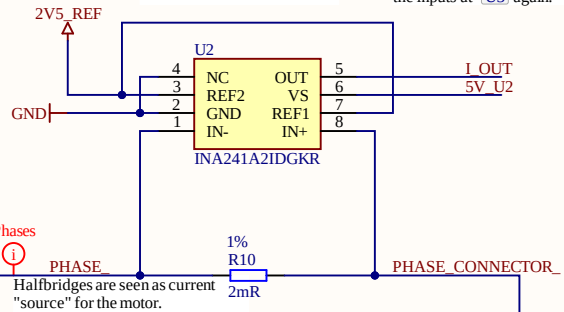
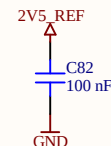


50A max @2mOhm = 1000mV
With fixed 20V/V Gain and 2.5V ref ->
0.5-4.5V bipolar single-ended

Phase Current Measurement + Single->Diff OPAMP



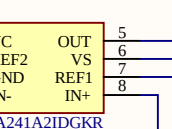
U2 the terminals IN- and IN+ are connected inverted. This is done to improve the PCB-layout. This is fixed, by inverting the inputs at U3 again.



Phases

PHASE

Halfbridges are seen as current "source" for the motor.



I_OUT

5V_U2

OUT

VS

REF1

IN+

IN-

REF2

NC

U2

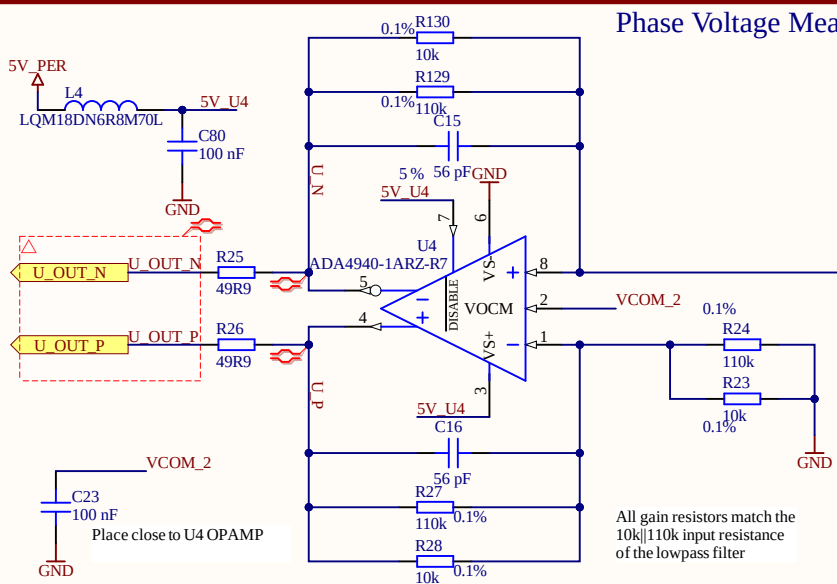
INA241A2IDGKR

1% R10

2mR

PHASE_CONNECTOR

Phase Voltage Measurement + Single->Diff OPAMP



Voltage measurement scaled with voltage divider R22 and R21:
Max 48V (LowPass max 60V as buffer)
10k||110k->
48V*10k/(10k+110k)=4V
48V/(10k||110k)=5.2mA

Max operation 60V:
60V*10k/(10k+110k)=5V
60V/(10k||110k)=6.5mA

LowPass Filter:
Max Target:
6000rpm at 18 pole pairs
->fg=1800Hz
With LowPass & Feedback
Lowpass. Resulting frequency
response calculated via LTSpice
Simulation ->1.745kHz

All gain resistors match the 10k||110k input resistance of the lowpass filter

Title	Phase_Measurements.SchDoc
Revision:	Rev04
Design Engineer:	K.Patel/D. Hufnagel
Project:	UZ_D_Inverter.PrjPCB

UltraZohm
www.ultrazohm.com
Date: 1/3/2025
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